

Fair Testing and Recording

A fair test, careful notes, and repeating to be sure — the habits that turn a guess into a reliable result.

Making a test fair and trustworthy

A **fair test** is one where only the changed variable is different and everything else is kept the same. This is the heart of **systematic investigation**.

The habits of a good investigator

- **Change one thing at a time** — keep all other conditions identical.
- **Keep careful notes** of everything you see and sense — did the oil splatter, smell, or smoke?
- **Repeat** the experiment to check your result is not a one-off accident.
- **Ask new questions** — one round of results usually sparks the next investigation.

Honest science: the textbook reminds us that even the simple puffing of a puri is *not yet fully understood* by scientists. It's fine for an investigation to end with new questions!

Why record observations?

Notes let you compare runs, spot patterns, and let others repeat your work. A result others can reproduce is far more trustworthy.

Tip: after an experiment, write down both what you expected and what actually happened — surprises are where new science begins.

Where students lose marks

Trap 1 — "Fair test" means equal/just. No — in science a fair test means only one variable changes; all else is kept the same.

Trap 2 — One trial is enough. A single result may be an accident. Repeating makes the conclusion reliable.

Trap 3 — Recording only "nice" results. Record everything, including splatter, smell and smoke — they can be important clues.

Model answers — write them like this

Example A. To compare boiling-hot, hot, and warm oil fairly, what must be kept the same?

Step 1: Identify the changed variable = oil temperature.

Step 2: Everything else must be constant.

∴ Use dough circles of the same thickness and size, dropped the same way each time.

Example B. Why repeat an experiment three times instead of once?

Step 1: A single trial could be affected by chance/error.

Step 2: Repeats that agree show the result is reliable.

∴ Repeating checks the result is not a one-off accident.

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